

QuantumXpo

VOID-NAV: Quantum-Assisted Navigation for GPS-Denied Drones

PROBLEM STATEMENT

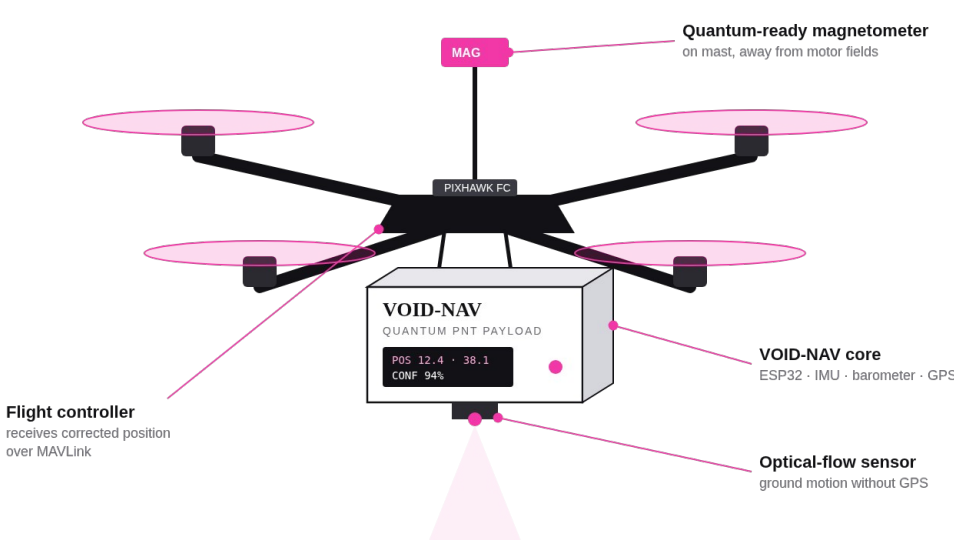
When GPS is jammed, blocked by terrain or simply unavailable, a drone falls back on its inertial sensors (IMU). Their small errors add up, and the position **drifts without bound**. Nothing on board can tell by how much.

Who is affected: defence UAVs, loitering systems, border and coastal patrol, and any drone flying in tunnels, forests or contested airspace.

Why current approaches fall short: low-cost MEMS IMUs drift hundreds of metres within minutes; navigation-grade INS and cold-atom sensors cost lakhs to crores; radar or camera fixes emit signals or fail in darkness and smoke.

PROPOSED SOLUTION

VOID-NAV is a small, passive navigation payload that mounts on a drone. It reads the Earth's **magnetic fingerprint**, matches it to a pre-built magnetic map with **quantum algorithms**, and corrects inertial drift. No satellites, no emissions.



- 1 • Map**
Survey once with GPS on: fingerprint [Bx, By, Bz, |B|, ∇B] per 2 m cell
- 2 • Sense**
GPS off: magnetometer, IMU, optical flow, barometer
- 3 • Match**
Quantum engine finds the matching map cell
- 4 • Correct**
Particle filter resets drift, sends position to autopilot

10^{-16} W

GPS power at the receiver: easy to lose

t^2

inertial error grows with time squared

UC-085

quantum sensing for GPS-denied navigation

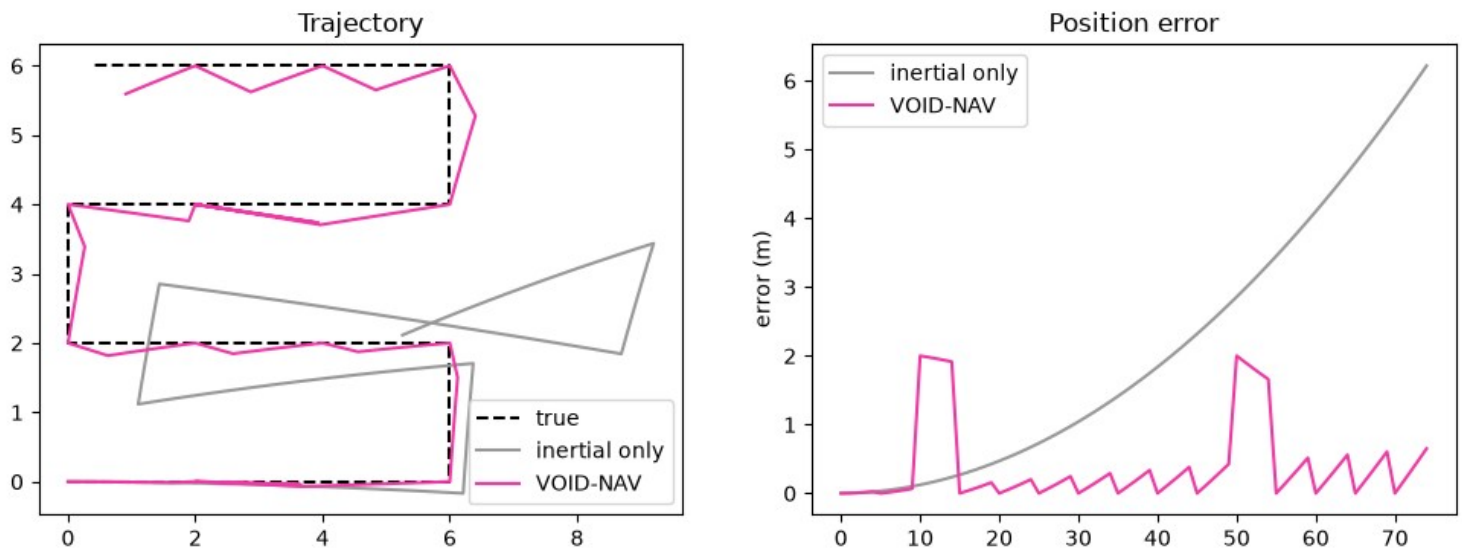
KEY FEATURES

- Full quantum pipeline**
sensing, search, learning and planning, each with a distinct job
- 5-D magnetic fingerprint**
matches field pattern and gradient, not a single value
- Grover + QSVM cross-check**
a correction is accepted only when both agree
- Quantum-sensor ready**
drop-in slot for an NV-diamond magnetometer
- Passive and retrofittable**
emits nothing; talks MAVLink to ArduPilot/PX4
- Low cost**
prototype under ₹4,000 in parts

QUANTUM SIMULATION

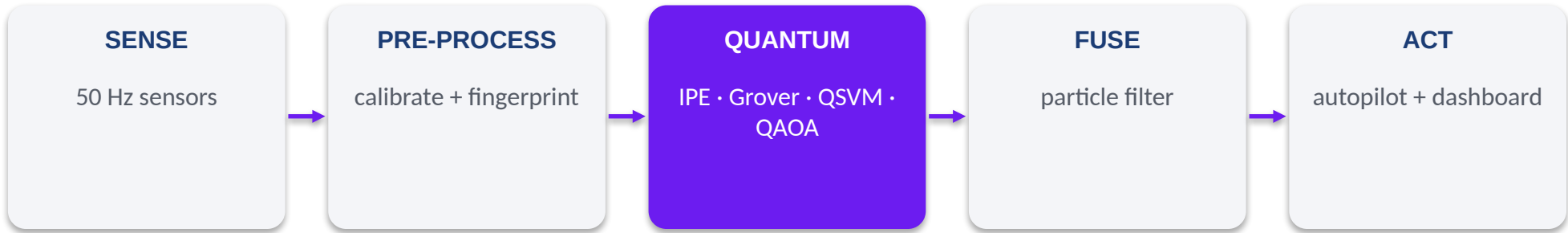
The full quantum pipeline runs in a Qiskit notebook on the **Qiskit Aer** simulator, on a 4x4 magnetic map. The simulation is part of the navigation loop: its outputs directly correct the drone's position.

Simulated block	What it computes	Verified result
Ramsey + IPE	field phase from an NV-diamond qubit	16/16 map cells recovered exactly (6-bit)
Grover (4 qubits)	matching map cell	96.1% of shots on correct cell 1011
QSVM (ZZ kernel)	zone: corridor / stairwell / lab	99.0% test accuracy
QAOA (p = 1)	best waypoint pair	matches brute-force optimum
Navigation loop	drift with vs without quantum fixes	mean error 2.10 m → 0.40 m (5.3×)



Navigation loop output from the notebook: inertial-only drift (grey) vs VOID-NAV with Grover + QSVM fixes (pink).

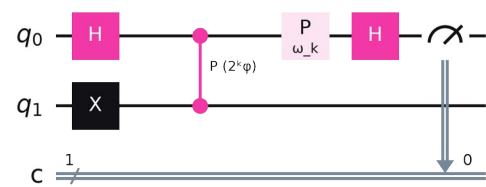
HOW IT WORKS



Sensor data streams in at 50 Hz. Every second the quantum engine takes a window of readings and returns candidate map cells; the particle filter accepts a correction only when **Grover and the QSVM agree**, then sends position + confidence to the autopilot. Quantum runs alongside flight control, never blocking it.

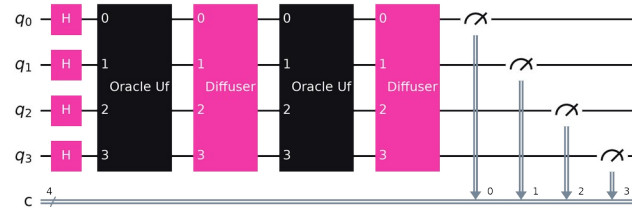
QUANTUM CIRCUIT / ALGORITHM

Q1 • Quantum sensing: Ramsey + Iterative Phase Estimation



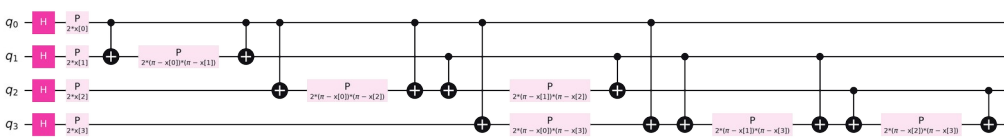
H creates superposition; the field imprints phase $\phi = 2\pi\gamma Bt$; phase kickback + IPE reads ϕ bit by bit.

Q2 • Grover's search: oracle + diffuser, 4 qubits



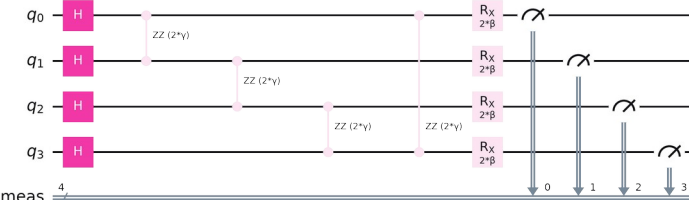
Oracle U_f flips matching cells; diffuser $2|s| - 1$ amplifies them; $\sim\sqrt{N}$ queries.

Q3 • Quantum-kernel SVM: ZZ feature map



$K(x, x') = |\langle \varphi(x) | \varphi(x') \rangle|^2$ from statevectors, fed to a classical SVM.

Q4 • QAOA route planning, p = 1



Cost Hamiltonian from a QUBO; mixer ΣX_i ; (γ, β) optimised on Aer.

IBM QUANTUM PLATFORM / TOOL

- Qiskit 2.x SDK (circuits, primitives)
- Qiskit Aer (AerSimulator)
- Qiskit quantum_info (Statevector kernels)
- Qiskit circuit library (zz_feature_map, DiagonalGate)
- IBM Quantum Platform via qiskit-ibm-runtime (hardware run planned)
- Also: scikit-learn, NumPy, Matplotlib, ArduPilot SITL

EXPECTED IMPACT

- GPS-independent, jam-proof positioning for defence drones
- Bounded error instead of unbounded drift (5.3× lower in simulation)
- Passive: no emissions to detect
- Low-cost payload that retrofits existing drones
- Dual use: disaster response, mining, tunnels, maritime

FUTURE SCOPE

- Run Grover and IPE on IBM Quantum hardware
- Replace the synthetic map with our real magnetometer survey
- Larger maps: more qubits, outdoor anomaly data
- Swap in a real NV-diamond quantum magnetometer
- Field trials with defence partners (DRDO / iDEX)